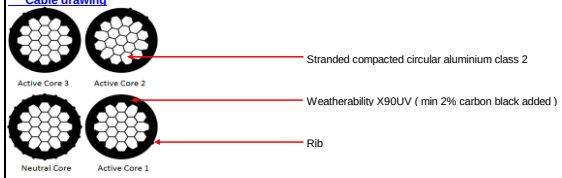
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Australia Pacific Electric Cables Pty Ltd
ABN 23 068 647 763

Cables Technical Specification - ABC (Aerial Bundled Cables 4 Cores) Aluminium Cl2 Power Cables XLPE (X90UV) Insulated AS/NZS 3560.1

*** Cable drawing**



Cable Configuration:
Al Class 2/X90UV 0.6/1kV

Cable Construction:
Conductor Material: Compacted aluminium Class 2 to AS/NZS 1125
Insulation: Weatherability X90UV (min 2% carbon black added)

Technical data

Item Description		Units	Average Values							
Product Code		-	ABC016A04	ABC025A04	ABC035A04	ABC050A04	ABC070A04	ABC095A04	ABC120A04	ABC150A04
Number of Cores		-	4 Cores (4C)							
form of conductors		-	Stranded compacted circular							
Total number of ribs	Neutral core	-	10	12	14	16	18	20	22	24
	Active core No. 1	-	1	1	1	1	1	1	1	1
	Active core No. 2	-	2	2	2	2	2	2	2	2
	Active core No. 3	-	3	3	3	3	3	3	3	3
Active / Neutral Cores	Conductor Cross Section Area (CSA)	mm^2	16	25	35	50	70	95	120	150
	Number of wires in conductor	mm	7	7	7	7	19	19	19	19
	Maximum Diameter of Wires	mm	1.70	2.20	2.58	2.96	2.16	2.52	2.86	3.18
	Diameter of conductor (minimum)	mm	4.5	5.8	6.8	8.0	9.6	11.3	12.8	14.1
	Diameter of conductor (maximum)	mm	4.8	6.1	7.2	8.4	10.1	11.9	13.5	14.9
	Maximum thickness of insulation at any point (excluding ribs)	mm	1.9	1.9	1.9	2.1	2.1	2.3	2.3	2.3
	* Minimum average thickness of insulation (excluding ribs)	mm	1.3	1.3	1.3	1.5	1.5	1.7	1.7	1.7
	* Minimum thickness of insulation at any point	mm	1.07	1.07	1.07	1.25	1.25	1.43	1.43	1.43
	Maximum diameter of core (excluding ribs)	mm	7.9	9.2	10.3	11.9	13.6	15.9	17.5	18.9
Maximum laid-up cores diameter		mm	19.1	22.2	24.9	28.7	32.8	38.4	42.2	45.6
Maximum overall diameter		mm	19.1	22.2	24.9	28.7	32.8	38.4	42.2	45.6
Approximate mass of cable (±10%)		kg/km	280	400	520	710	970	1350	1660	2020
Height of active core numbering		mm	≥ 3	≥ 3	≥ 3	≥ 5	≥ 5	≥ 5	≥ 5	≥ 5
Maximum d.c. resistance of conductor in the cable at 20°C		Ω/km	1.91	1.20	0.868	0.641	0.443	0.320	0.253	0.206
Maximum a.c. resistance of conductor in cable at 80°C		Ω/km	2.37	1.49	1.08	0.796	0.551	0.398	0.315	0.257
Positive sequence reactance of cable at 50Hz		Ω/km	0.106	0.102	0.0982	0.0924	0.0893	0.0868	0.0844	0.0844
Maximum continuous current carrying capacity per phase	Still air	A	44	59	72	88	110	136	157	179
	Wind speed of 1m/s	A	74	97	118	142	177	216	249	282
	Wind speed of 2m/s	A	86	113	137	165	207	257	300	343
Minimum breaking load of conductor		kN	2.2	3.5	4.9	7.0	9.8	13.3	16.8	21.0
Minimum breaking load of cable		kN	8.8	14.0	19.6	28.0	39.2	53.2	67.2	84.0
Minimum bending radius of single core		mm	30	40	40	50	60	70	70	80
Minimum bending radius of cable		mm	115	135	150	160	285	345	380	410
Highest value for maximum working tension		kN	2.46	3.92	5.49	7.48	11.0	14.9	18.8	23.5
Highest value for everyday tension		kN	1.58	2.52	3.53	5.00	7.10	9.60	12.1	15.1
Voltage drop at 50 Hz 80°C		mV/A.m	4.11	2.59	1.87	1.39	0.97	0.71	0.57	0.47
A.C. resistance at 50 Hz 90°C		Ω/km	2.45	1.54	1.11	0.822	0.569	0.411	0.325	0.265
Remark										

Core Identification:
Active core 1: 1 ONE (with 1 rib)
Active core 2: 2 TWO (with 2 ribs)
Active core 3: 3 THREE (with 3 ribs)
Neutral core: Evenly spaced ribs around

Colour Identification:
Insulation: Black

Operating Temperature:
Maximum Continuous Conductor Temperature: 80˚C
Emergency Conductor Temperature: 105˚C
Short Circuit (5 secs): 250˚C
Coefficient of linear thermal expansion: 23 x 10^-6^C

Insulation Markings:
Active Core 1: Numerals & letters (example: **1 ONE**) and meter marking
Active Core 2: Numerals & letters (example: **2 TWO**)
Active Core 3: Numerals & letters (example: **3 THREE**)

Polyethylene Terephthalate Tape (non-hygroscopic) Markings (on active core 1):
APEC YEAR X90UV ELECTRIC AERIAL CABLE 0.6/1 KV SIZE SQMM AL AS/NZS 5000.1

Electrical Characteristics:
Electrical characteristics are calculated based on ambient air temperature 40˚C
Votage Rating: 0.6/1kV
Conductors Spark Test (< 35mm²): 15 kV a.c. or 22kV d.c.
Conductors Spark Test (> 35mm²): 20 kV a.c. or 30 kV d.c.
High Voltage a.c. Test for 4 hours: 3.75kV
Minimum Insulation Resistance constant @20°C: 3000GΩ.m
Minimum Insulation Resistance constant @90°C: 3GΩ.m

Mechanical Characteristics:
Modulus of elasticity of conductor: 59GPa

Standard Compliance:

AS/NZS 5000.1	Part 1: For working voltages up to and including 0.6/1 (1.2kV)
AS/NZS 3808	Insulation and sheathing materials for electric cables
AS/NZS 3560.1:2000	Electric cables - Cross-linked polyethylene insulated - Aerial bundled - For working voltages up to and including 0.6/1 (1.2kV)
AS/NZS 3008.1.1:2009	Electrical installations - Selection of cables - Cables for alternating voltages up to and including 0.6/1 kV - Typical Australian installation conditions
AS 2848.1	Aluminium and aluminium alloys - Compositions and designations - Wrought products
AS/NZS 1660	Test methods for electric cables, cords and conductors
AS 1531	Conductors - Bare overhead - Aluminium and aluminium alloy
AS/NZS 1125	Conductors in insulated electric cables and flexible cords
RoHS	Restriction of Hazardous Substances

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